Data
Science

Defect Prediction Using Logistic Regression

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- **PROJECT TYPE**
  IVY Data Science Internship Project

2026

Business Problem & Objective

Business Problem :

The manufacturing process is experiencing defective products, leading to increased costs, inefficiencies and quality issues.

Objective:

- Build a predictive model using historical data
- Enable early detection of product defects
- Improve overall product quality
- Reduce manufacturing costs and waste

Data Overview & Consolidation

01

ERP, MES, QMS, SCADA

- production event data
- Same structure
- Contain machine details, operator details, process parameters, and defect flag

02

Machine Master Data

machine configuration and capability

03

Operator Master Data

operator role and experience .

- Combined all production datasets into one unified dataset
- Merged machine and operator master data with production data
- Final dataset includes:
  - Operational data
  - Machine details
  - Operator information
  - Quality outcome (defect_flag)

Feature Preparation & Analysis

MISSING VALUE TREATMENT

- 19 out of 38 columns contained missing values.
- Numerical columns were imputed using median values to handle skewed distributions.
- Categorical columns were filled using mode imputation to preserve the most frequent category.

OUTLIER HANDLING

- Outliers were detected using the Interquartile Range (IQR) method.
- Values beyond 1.5×IQR from Q1 and Q3 were identified as outliers.
- Extreme values were capped at threshold limits to reduce their impact on model performance.

DATA QUALITY ASSURANCE

- After preprocessing, the dataset was validated for consistency and completeness.
- Feature scaling and encoding were applied where necessary.
- The cleaned dataset is now ready for logistic regression model training.

Results & Validation

Confusion Matrix

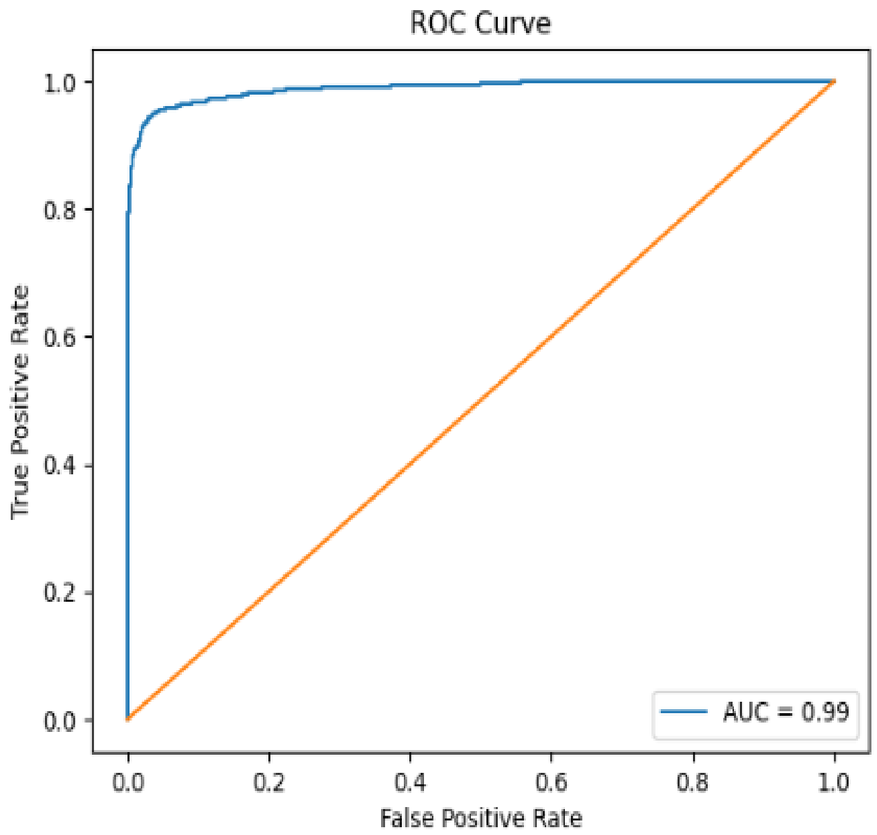
	Predicted Non-Defect	Predicted Defect
Actual Non-Defect	1716 True Negative	69 False Positive
Actual Defect	33 False Negative	623 True Positive

Accuracy	96%
Precision (Defect Class - 1)	90%
Recall (Defect Class - 1)	95%
F1-score (Defect Class - 1)	92%

- With 96% accuracy, the model reliably identifies product quality outcomes.
- The 90% precision score indicates minimal false alarms, reducing unnecessary interventions.
- The 95% recall rate ensures that the vast majority of actual defects are successfully detected.
- The balanced F1-score of 92% confirms robust overall model performance.

Total Matches: 2339 | Mismatches: 102

Model Performance – ROC Curve



AUC Score: 0.99

Model Interpretation:

- The ROC curve represents the relationship between:
  - True Positive Rate (Recall)
  - False Positive Rate
- A curve closer to the top-left corner indicates strong model performance
- The model shows a high probability of correctly classifying both classes across different thresholds

Model Comparison

01

Logistic Regression

96% accuracy with high interpretability. Selected as the primary model for deployment.

02

Decision Tree

Captures non-linear patterns in data. Provides visual decision rules for analysis.

03

Random Forest

More robust ensemble method. Reduces overfitting through multiple trees.

Final Selection:

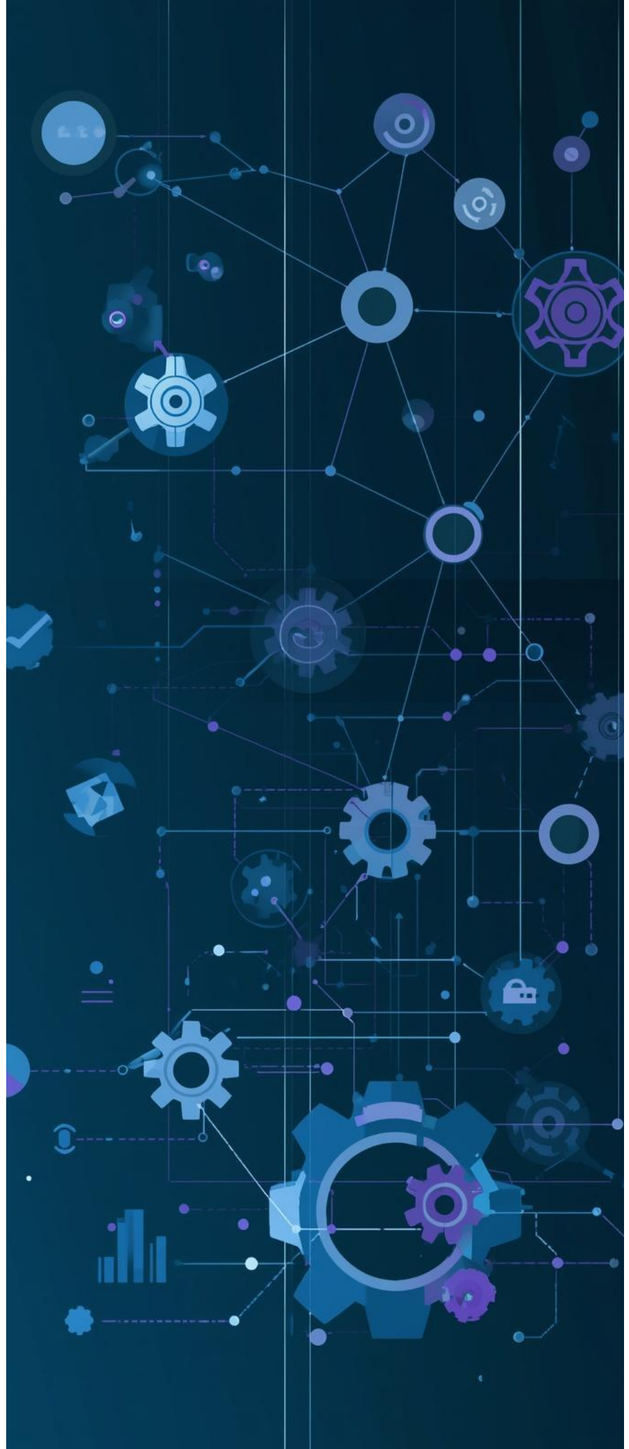
Logistic Regression chosen for optimal balance of accuracy and interpretability.

Recommendations

STRATEGIC ACTIONS FOR MODEL DEPLOYMENT

- Deploy the logistic regression model for early defect detection in production lines.
- Reduce manufacturing costs and improve overall product quality through predictive insights.
- High recall (95%) ensures most defects are caught, minimizing missed defective products.
- Explore Random Forest algorithm for future improvement and enhanced model robustness.

Thank You!



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Thank you for your attention. This defect prediction project using logistic regression demonstrates how data-driven approaches can improve manufacturing quality and reduce costs.

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